# Computer Networks

PPT-1: Overview of Computer Networks

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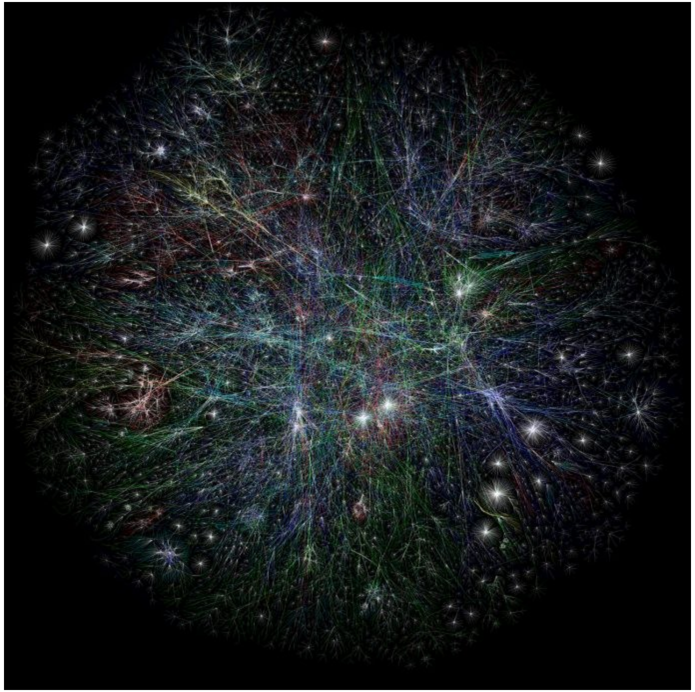
# What is a computer network

A computer network is a set of interconnected computers that communicate with each other and share resources.

The purpose of a computer
network is to enable the exchange of information and resources among different devices.



Ref. Partial map of the Internet based on the January 15, 2005 data found on opte.org
Computer Network Wikipedia

# Characteristics of Computer Network

1. Delivery : Sending to correct destination
2. Accuracy : No mismatch in Data / Error Free
3. Timeliness : It should reachin proper time
4. Jitter : Variation in packet delivery delay

# Features of a Computer Network

1. Connectivity
2. Scalability
3. Reliability - fault tolerance
4. Performance: *Throughput & delay*
5. Security
6. Interoperability
7. Manageability
8. QoS
9. Flexibility
10. Distributed Processing
11. Resource Sharing
12. Accessibility
13. Cost-Effectiveness
14. Adaptability
15. Standardization

Performance can be measured in many ways, including transit time and response time. Transit time is the amount of time required for a message to travel from one device to another. Response time is the elapsed time between an inquiry and a response.

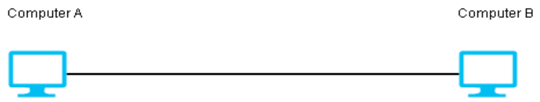
# Key components

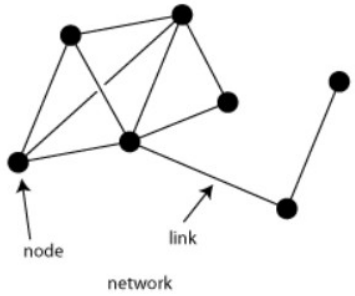
**Message:** Information that is to be transmitted. E.g. Text, Numbers, Pictures, Audio , Video etc.

**Nodes/Devices**: These are the individual computers, servers, printers, routers, and other devices that are connected to the network.

**Communication Links**: These are the physical or wireless connections that allow data to be transmitted between nodes.

**Protocol** : Set of Rules that govern the data communication.



# Data Representation

1. **Text** : Represented as bit patterns or sequence of bits ( 1100110)
2. **Numbers** : Bit Pattern
3. **Images** : Matrix of pixels
    a. Black or white (0/1)
    b. Gray color (0-255)
    c. Color Image 3 Matrices, one for each color (RGB)
4. **Audio**: voice , recording, music -> Analog to digital
5. **VIdeo :** Sequence of pictures which may include audio as well contained in a single file

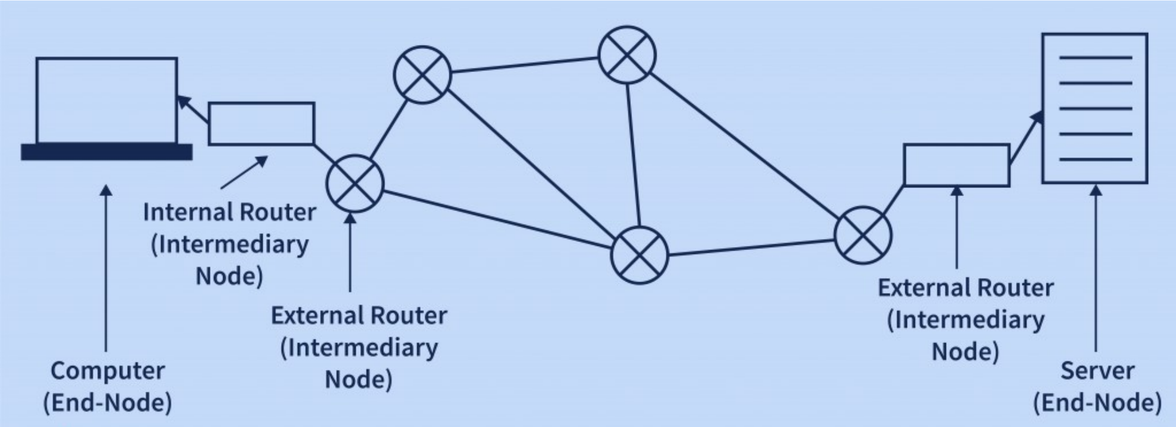
# Typical Elements of a Network

Node

- End Devices
- Intermediary devices

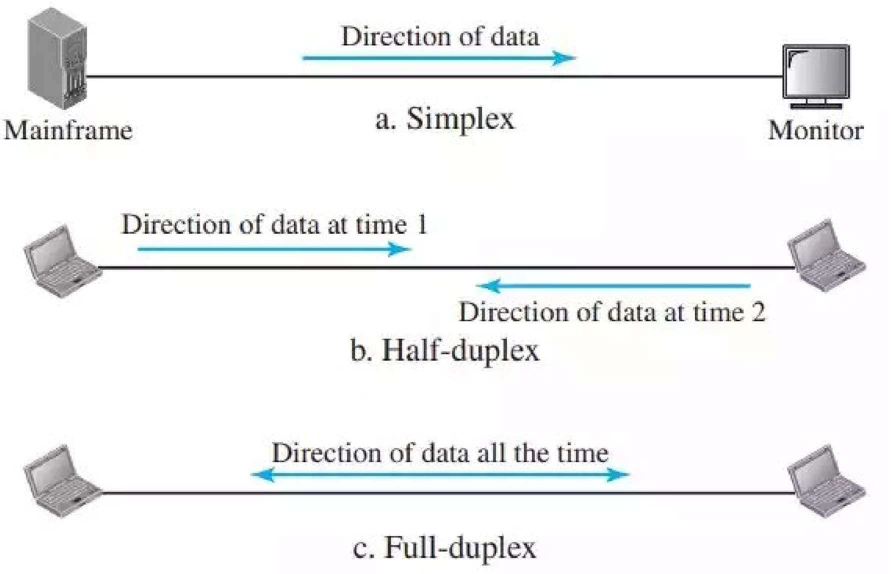
Types of communication links

- Wired
- Wireless



# Transmission Modes

a. **Simplex**
    - TV Transmission
    - Keyboard
    - Monitor
b. **Half Duplex**
    - Walkie talkie
c. **Full Duplex**
    - Telephone

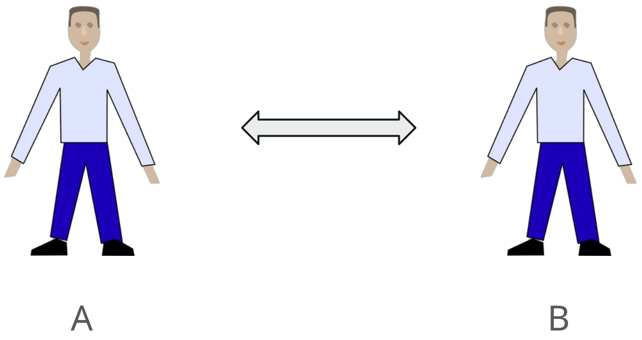


# Protocols

A protocol is a set of rules and conventions that govern how data is exchanged and transmitted between devices within a network.  They ensure that devices from different manufacturers or running different software can communicate effectively.

Protocols define:

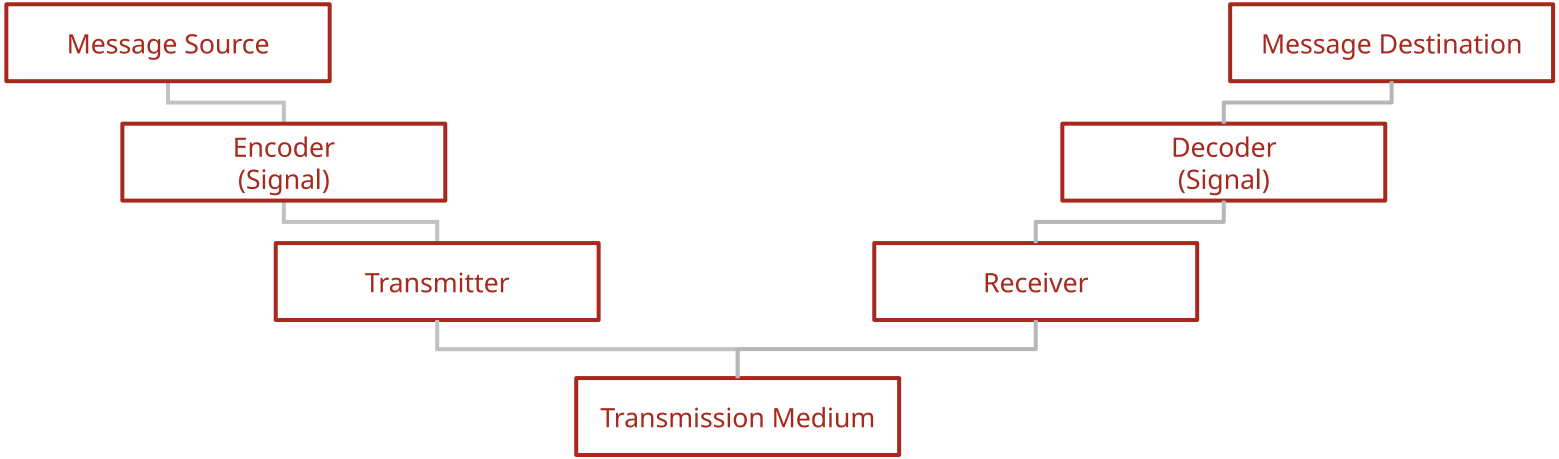
- Format
- Timing
- Sequencing
- Error control



# Elements of Protocol

1. Message Encoding
2. Message formatting and encapsulation
3. Message timing
4. Message size
5. Message Delivery Options

# Message Encoding



# Message Formatting

- Agreed Format
- Encapsulate to identify sender and receiver
  - Source and Destination address

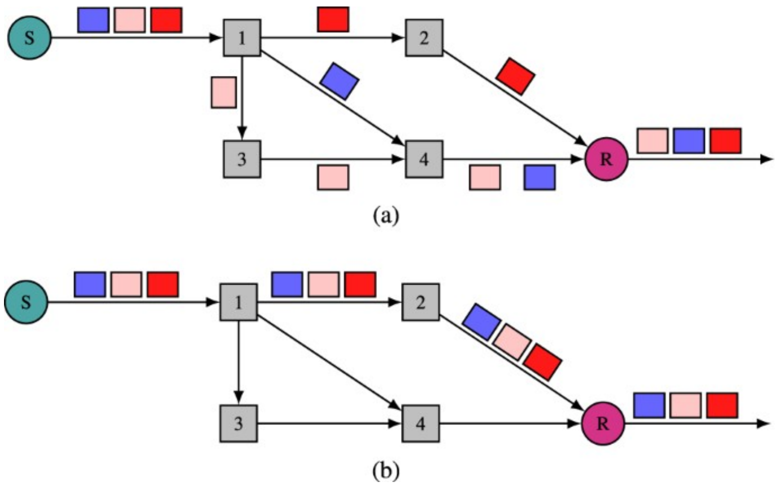
# Message Timing

- Flow Control
  - Slow Receiver
  - High speed Sender

- Response Timeout

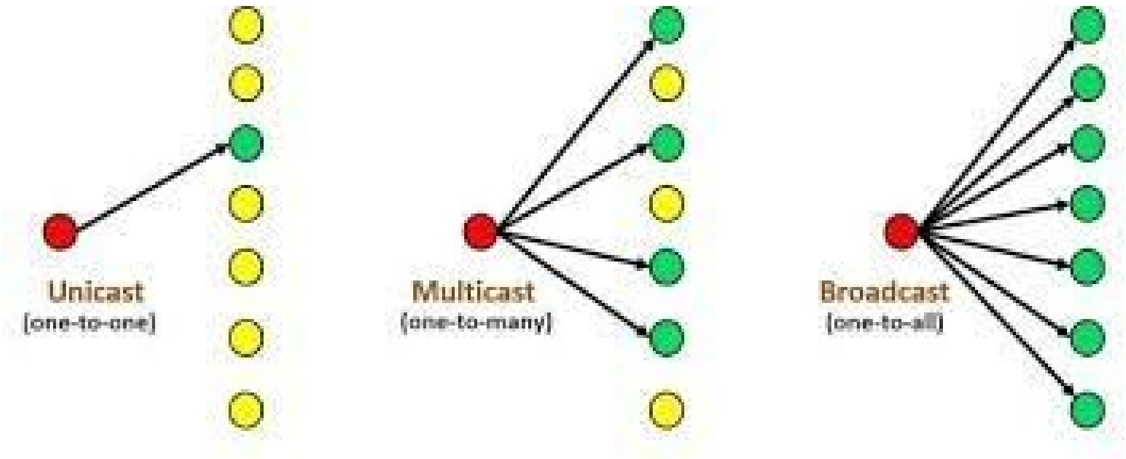
# Message Size

- Long messages must be broken into smaller pieces

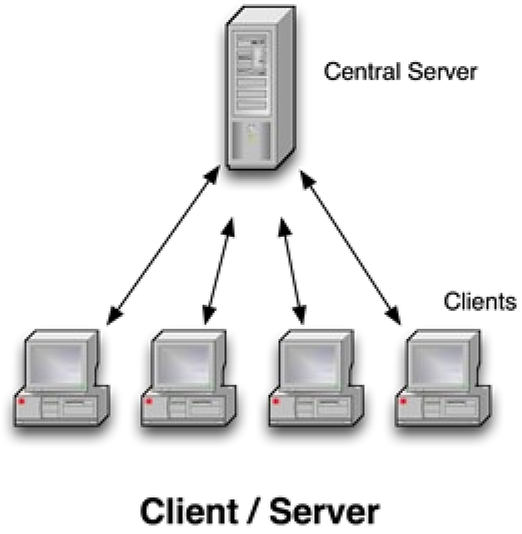


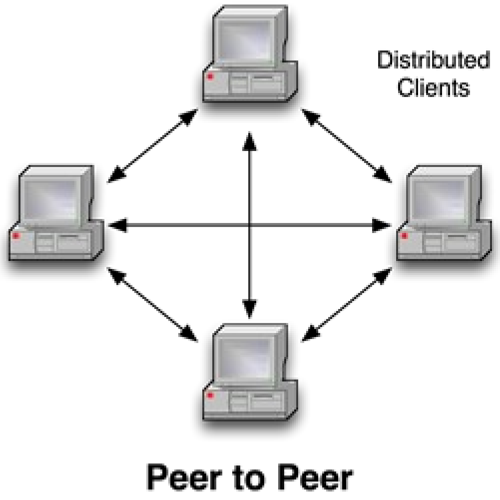
# Message Delivery Options

- Unicast
- Multicast
- Broadcast



# Network Architecture / Models



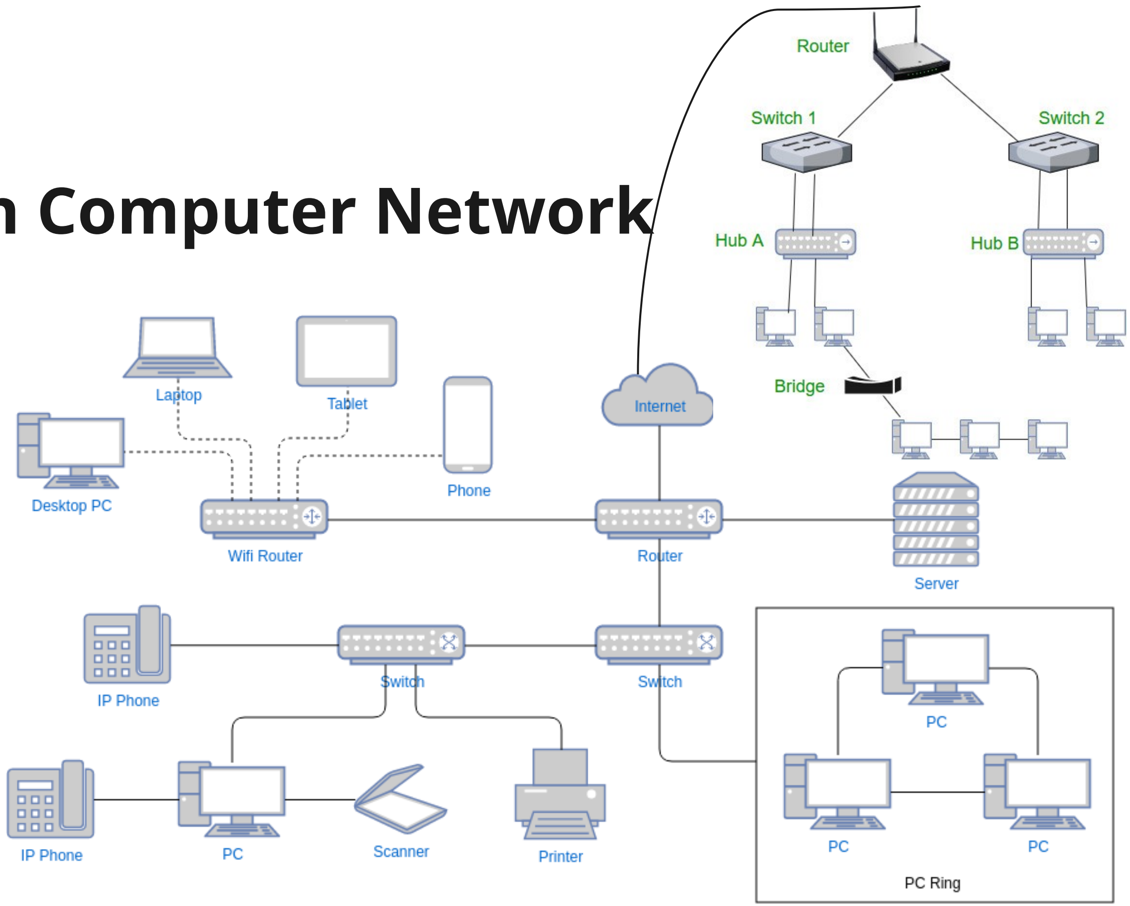
# Equipment used in Computer Network

**End Nodes**

- Computers
- Network Printers
- VOIP Phones
- Network Security Cameras
- Mobile Devices

**Intermediary Devices**

- Repeater
- Hub
- Bridge
- Switch
- Router
- Access Point
- Modem
- Firewall
- Gateway



# Equipment used in Computer Network

Communication Link (Media)

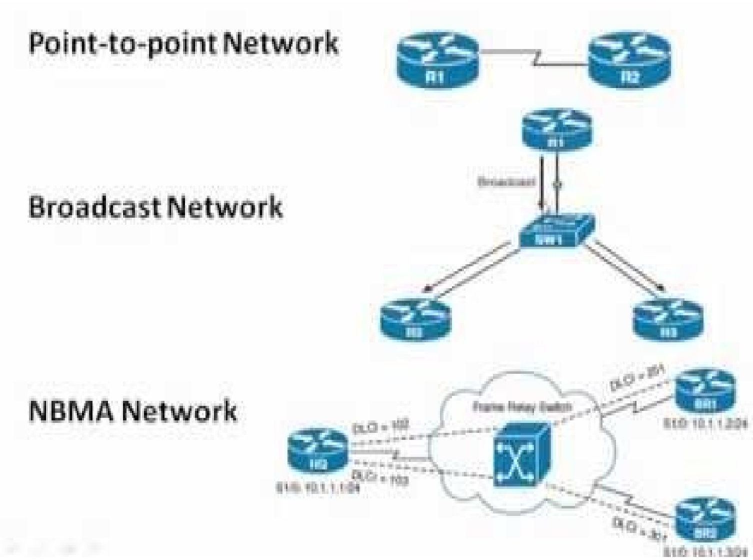
- Wired or guided media
  - Ethernet straight through cable: *for connecting different kind of devices*
  - Ethernet Crossover cable: *for connecting same kind of devices*
  - Fiber optic cable
  - Co-axial cables
  - USB Cables

- Wireless or unguided media
  - **Infrared** : *TV Remote, AC remote ,etc…*
  - **Radio Waves**: *Wifi,  Bluetooth*
  - **Microwaves**: *Cellular System, Satellite link*

# Network Taxonomy

1. Transmission Technology
   1. Point to Point Network
   2. Broadcast Network
   3. Non Broadcast Multi Access Network

2. Scale
   4. PAN
   5. LAN
   6. MAN
   7. WAN
   8. INTERNET



# Classification on the basis of Scale

| Parameters | LAN | MAN | WAN |
|---|---|---|---|
| **Ownership of Network** | Private | Private / Public | Private / Public |
| **Geographical Area** | 1KM to 10 KM | 10 KM to 100 KM | Beyond 100 KM |
| **Design and maintenance** | Easy | Difficult | Difficult |
| **Propagation Delay** | Short | Moderate | Long |
| **Speed** | High | Moderate | Low |
| **Fault Tolerance** | More Tolerant | Less Tolerant | Less Tolerant |
| **Congestion** | Less | More | More |
| **Used for** | College, School, Hospital. | Small towns, City. | Country/Continent. |

TABLE 2 Comparison of Characteristics of PAN, LAN, MAN, CAN, WAN

| Sr.No. | Parameters | PAN | LAN | MAN | CAN | WAN |
|---|---|---|---|---|---|---|
| 1. | Ownership | Private | Private | Private or Public | Private | Private or Public or Leased |
| 2. | Design Maintenance | Easy | Easy | Difficult | Difficult | Difficult |
| 3. | Speed | Low | High | Moderate | High | Low |
| 4. | Congestion | Less | Less | More | Medium(More than PAN) | More |
| 5. | Area Covered | Short up to 10m | Confined to single building such as school, office etc. | Covers up to entire city. | 1Km to 5 Km | large area covering country even whole world |
| 6. | Technology Used | | Ethernet, Token Ring | | Ethernet, Token Ring | ATM, Frame Delay X.25 |
| 7. | Propagation Delay | Short | Short | Moderate | | Long |
| 8. | Communication Medium | Wireless or Wired | Coaxial Cable | Optical Fiber, Coaxial Cable or wireless | Optical Fiber | Satellite Links |
| 9. | Bandwidth | | High | Moderate | High | Low |
| 10. | Cost | Cheap | Inexpensive | Cost Effective | Cheap | Expensive |
| 11. | Fault Tolerance | | More | Less | | Less |
| 12. | Used For | Revolves around single person | Home, Office School | Cable TV Network | Enterprise, University, Government Institution | Banks Branches in a country. |

# Next Lecture

Network Topology